

User Flow & System Architecture

Grounded AI Copilot, Human-in-the-Loop Governance & Modular Engineering Workspace

Grounded RAG

Hybrid PostgreSQL full-text search (`tsvector`) and vector embeddings (`pgvector`) with strict citation verification.

Zero-Trust AI

AI outputs are untrusted drafts. All domain mutations require explicit human review and atomic transactional commits.

Traceable Core

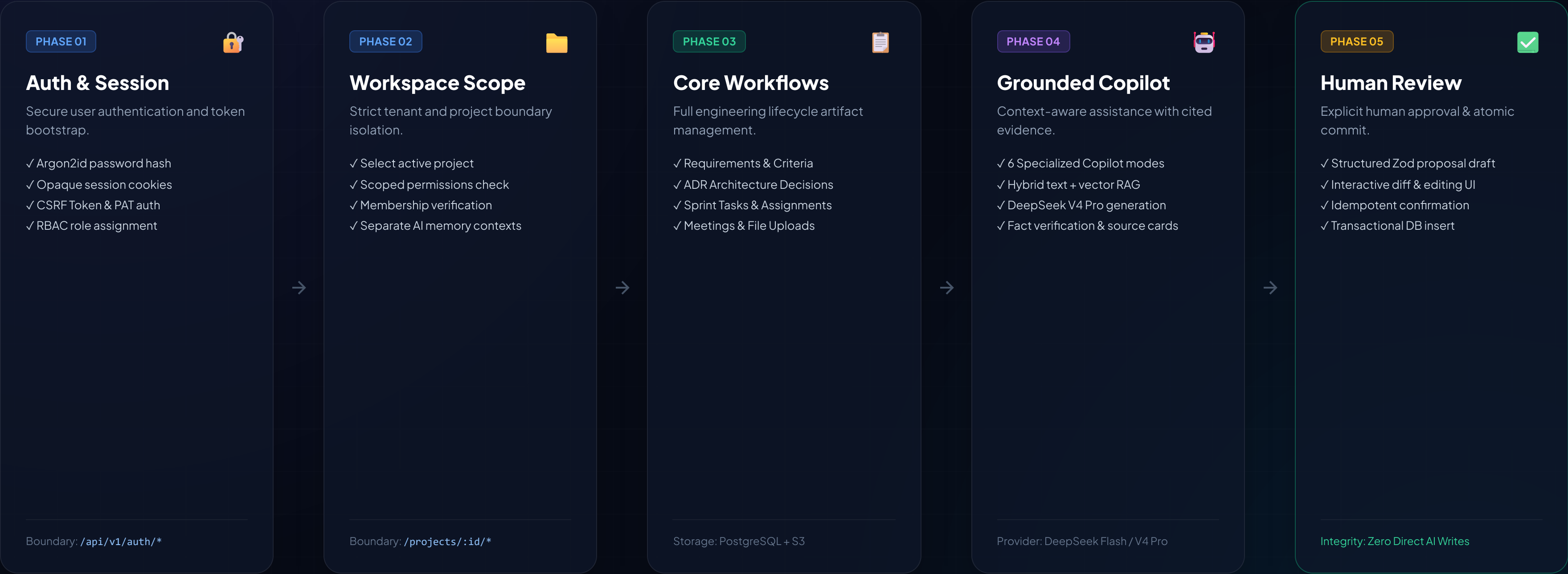
Bi-directional relations connecting Requirements, Acceptance Criteria, ADR Decisions, Tasks, and Meetings.

MCP Agent Ready

Native Model Context Protocol server allowing external IDE agents (Cursor, Claude) to query and update workspace context.

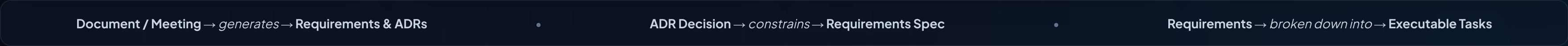
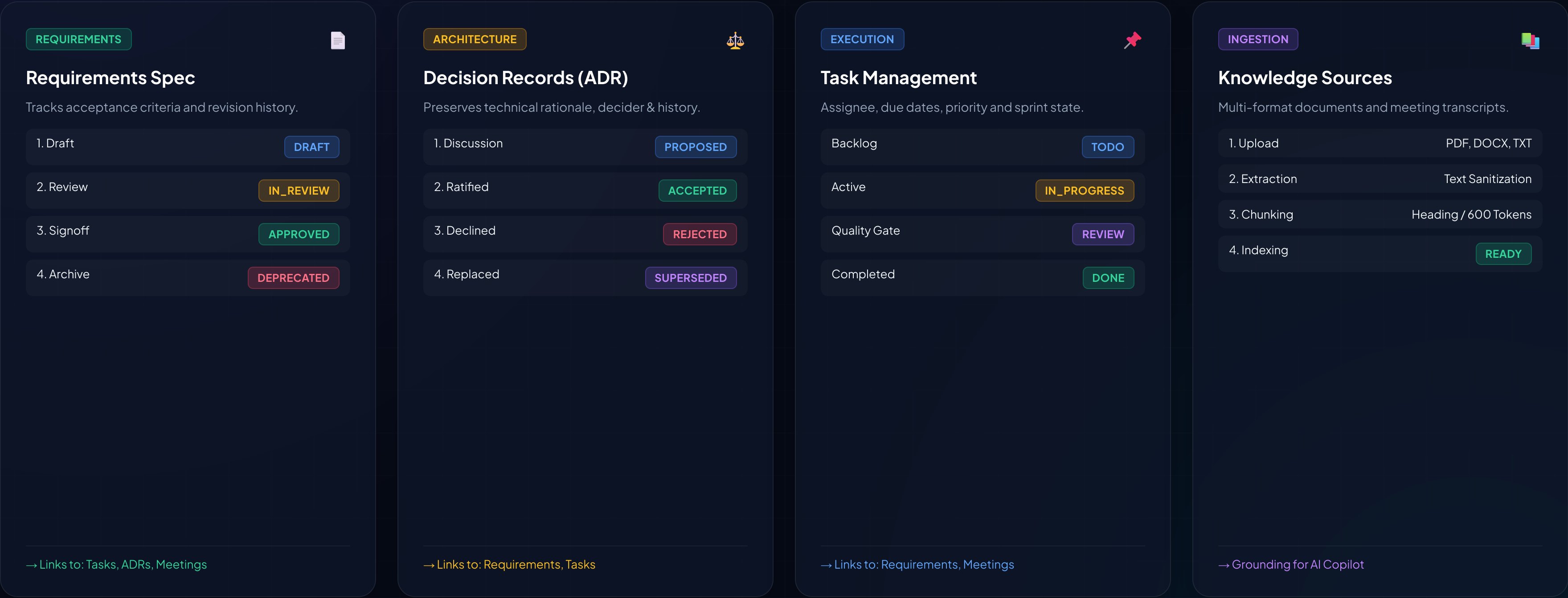
End-to-End User Journey

From authenticated session to grounded AI collaboration and transactional execution



Core Entity Lifecycle & Interconnection

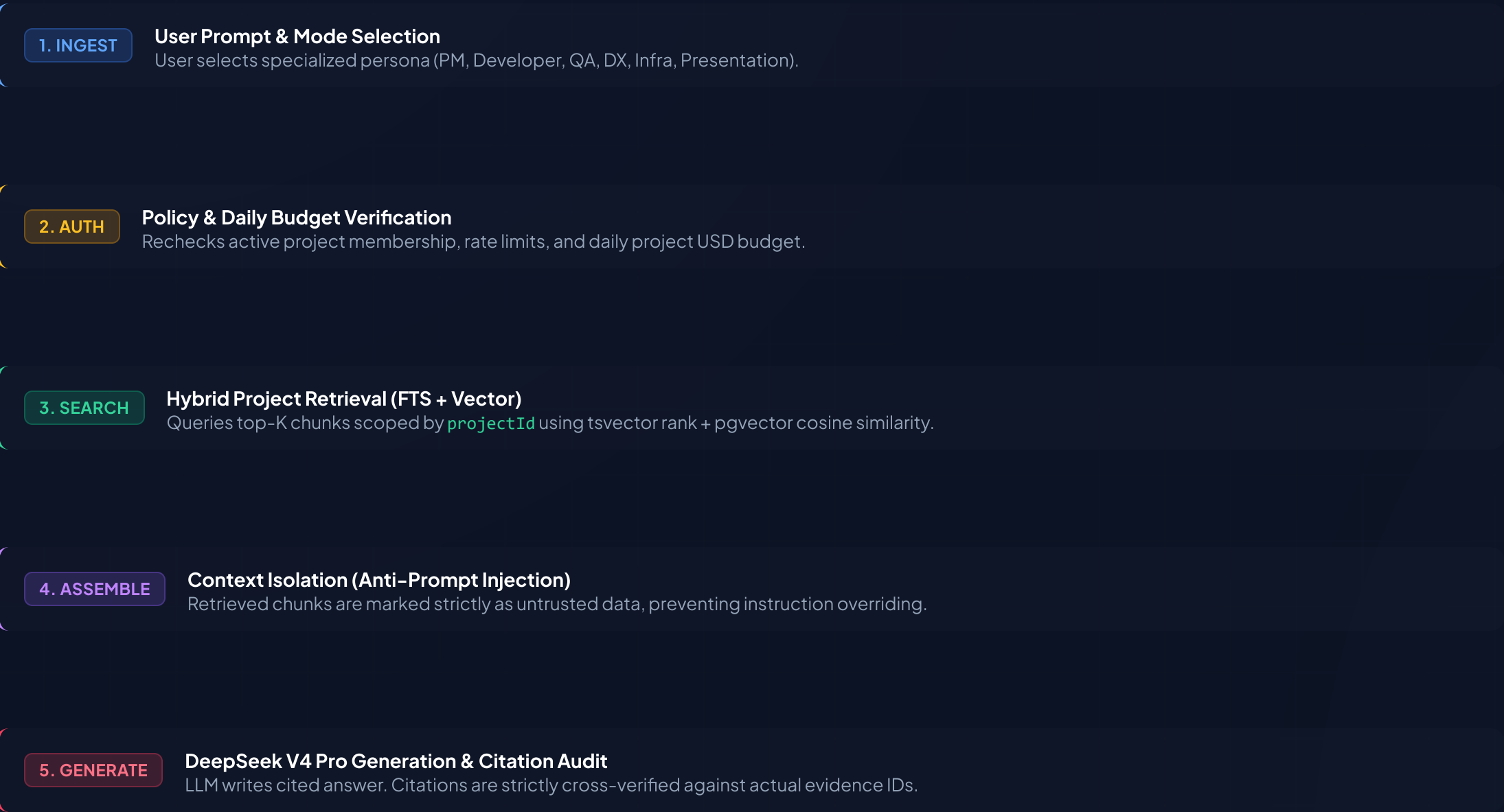
Bi-directional traceability linking documents, meetings, requirements, decisions, and tasks



Grounded AI Copilot & RAG Pipeline

Guaranteed factuality: Permission-aware search, anti-injection framing, and citation verification

⚡ End-to-End Query & Citation Lifecycle

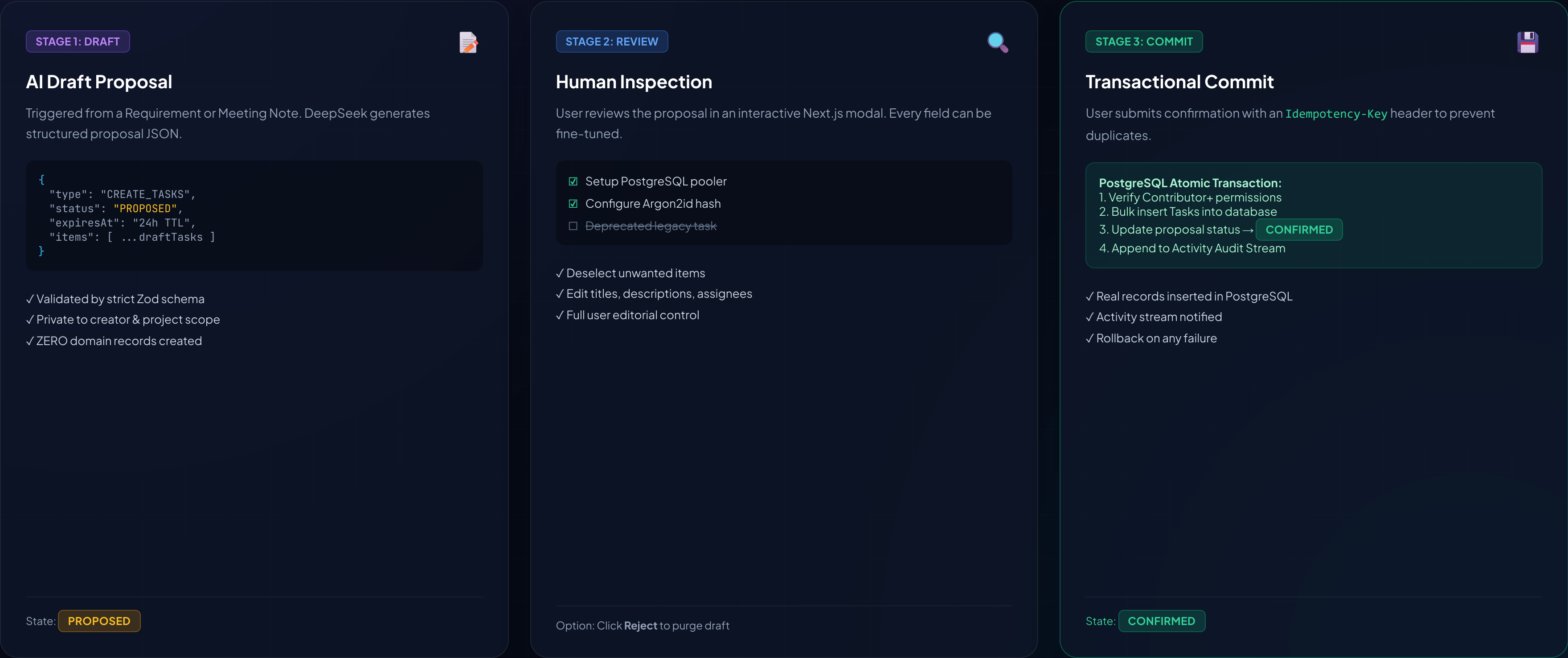


6 Operational Copilot Modes



Human-in-the-Loop Proposal Lifecycle

AI outputs structured drafts; humans inspect, edit, and transactionally commit domain entities



Model Context Protocol (MCP) Integration

Connecting external AI coding agents (Cursor, Claude Desktop, Antigravity) to production workspace data

 External AI Integration Stack

1. External Agent Layer

Developers code in their favorite IDE while maintaining live workspace sync.

Cursor / Claude / Antigravity

2. Transport & Authentication

Streamable HTTP / SSE transport secured by Personal Access Tokens (`aiw_pat_*`).

PAT Bearer Token

3. Project Authorization Engine

Validates user role and project scope on every single MCP tool invocation.

RBAC Guard

✓ Production Deployed: Connected directly to live Render backend & Supabase PostgreSQL.

 Registered Workspace MCP Tools

`list_tasks`

Query tasks by status, assignee, priority, or milestone.

`get_dashboard`

Retrieve real-time project metrics, progress %, and overdue counts.

`list_requirements`

Fetch active requirements and criteria specifications.

`list_decisions`

Access ratified ADR architectural decisions and rationales.

`generate_task_proposal`

Trigger structured AI breakdown proposals for requirements.

`search_workspace`

Hybrid full-text and semantic knowledge search across all records.

// Live Agent Execution Result
mcp.call_tool("get_dashboard") → { progress: 91%, totalTasks: 11, activeMembers: 3 }

Enterprise Role–Based Access Control (RBAC)

Granular capability boundaries enforced at every API endpoint and database query

Workspace Capability	👑 OWNER	🛡️ MAINTAINER	✏️ CONTRIBUTOR	👁️ VIEWER
View Dashboard, Activity Feed & Knowledge Records	ALLOWED	ALLOWED	ALLOWED	ALLOWED
Query Grounded AI Copilot & Hybrid Search	ALLOWED	ALLOWED	ALLOWED	ALLOWED
Create & Edit Requirements, ADRs, and Meetings	ALLOWED	ALLOWED	ALLOWED	DENIED
Create, Move & Complete Tasks	ALLOWED	ALLOWED	ALLOWED	DENIED
Review, Edit & Confirm AI Proposals	ALLOWED	ALLOWED	ALLOWED	DENIED
Upload Documents & Trigger Background Indexing	ALLOWED	ALLOWED	ALLOWED	DENIED
Manage Integrations (GitHub, PATs) & Webhooks	ALLOWED	ALLOWED	DENIED	DENIED
Invite, Modify & Remove Project Members	ALLOWED	ALLOWED	DENIED	DENIED
Transfer Ownership & Archive Project	ALLOWED	DENIED	DENIED	DENIED

🔒 **Mandatory Scope Joins:**
Every SQL query filters on both `projectId` and `actorId`.

🛡️ **Revocable Server Sessions:**
Sessions stored in DB; instant revocation without JWT leakage risks.

⚡ **AI Non–Elevation Rule:**
AI prompts inherit exact user credentials; AI modes never bypass permissions.